Band nonoverlap from division by a strip Blaschke factor

AI-assisted research draft for review

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Theorem 1 (Graph-bottom water waves)**.** *Let $h > 0$, and let b be real, 2π-periodic and C^1, with $\min_x(h - b(x)) > 0$. For the exact harmonic-extension Dirichlet–Neumann operator with a flat top and homogeneous Neumann bottom condition, its ordered Bloch eigenvalues satisfy*

$$\Lambda_n(\theta_a) \le \Lambda_{n+1}(\theta_b) \quad \text{for every } n \ge 0 \text{ and every } \theta_a, \theta_b \in \mathbb{R}. \tag{1}$$

Consequently $\max_\theta \Lambda_n(\theta) \le \min_\theta \Lambda_{n+1}(\theta)$ for every n. Eigenvalues are counted with multiplicity, and touching bands are allowed.

The proof first establishes a weighted strip theorem. Its crucial geometric condition is a positive weight on the middle of a doubled strip. An arbitrary positive surface weight does not suffice for this argument.

1. Weighted strip and its analytic representation

Fix $P = 2\pi$ and $d > 0$. Let p be a continuous real periodic function with $\min p > 0$, and let

$$\rho_k = \frac{p_k}{\cosh(dk)}, \qquad k \in \mathbb{Z}. \tag{2}$$

Then ρ is continuous and positive. Indeed it is the value on the middle line of the harmonic function on $|\operatorname{Im} z| < d$ with boundary value p on both edges, so $\min p \le \rho \le \max p$ by the maximum principle. Consider the harmonic extension in the single strip $-d < t < 0$, with zero Neumann data at $t = -d$, and use top mass $\int_0^P \rho(s)|u(s)|^2 ds$. All quasiperiodic functions below have unitary character $e^{iP\theta}$. The strip form has the explicit Fourier value

$$q_\theta[u] = P \sum_m (m + \theta) \tanh(d(m + \theta))|u_m|^2.$$

Consequently $q_\theta + \int \rho|u|^2$ is equivalent to the $H^{1/2}$ norm squared. The form is closed, and the compact embedding $H^{1/2} \hookrightarrow L^2(\rho ds)$ gives compact resolvent. Its ordered min–max formula can also be seen directly from an orthonormal eigenbasis: an $(n+1)$-dimensional subspace contains a nonzero vector orthogonal to the first n basis vectors, while the span of the first $n+1$ attains the opposite inequality.

For a top trace $u(s)=\sum_m u_m e^{i(m+\theta)s}\in H^{1/2}$ define its bottom trace and holomorphic continuation by

$$v(s)=\sum_m \frac{u_m}{\cosh(d(m+\theta))}e^{i(m+\theta)s},\qquad V(z)=\sum_m \frac{u_m}{\cosh(d(m+\theta))}e^{i(m+\theta)z}. \tag{3}$$

Thus V is holomorphic on $\mathcal{D}_d=\{|\operatorname{Im} z|<d\}$, has the same character, belongs to H^1 on one doubled cell, and its two edge traces belong to $H^{1/2}$. The original harmonic extension, with $r=t+d\in[0,d]$, is

$$U(s,-d+r)=\frac{V(s+ir)+V(s-ir)}{2}. \tag{4}$$

Conversely every holomorphic H^1 function on a doubled cell with unitary character gives an admissible top trace by (4). These claims follow directly from Fourier series: the multipliers on either edge in (3) are bounded, and the analytic Dirichlet energy weights a coefficient v_m by $(m+\theta)\sinh(2d(m+\theta))$, with its nonnegative limiting interpretation at zero.

Write $V_\pm(s)=V(s\pm id)$ and $V_0(s)=V(s)$. The strip energy and weighted top mass are respectively

$$Q(V)=\frac{1}{2}\int_{(0,P)\times(-d,d)}|V'(z)|^2\,ds\,dt, \tag{5}$$

$$M(V)=\frac{1}{4}\int_0^P \rho(s)\big(|V_+(s)|^2+|V_-(s)|^2\big)\,ds+\frac{1}{2}\int_0^P p(s)|V_0(s)|^2\,ds. \tag{6}$$

For (5), differentiate (4) and use $|a+b|^2+|a-b|^2=2|a|^2+2|b|^2$. For (6), expand the top square in (4). The real part of its cross term is

$$\operatorname{Re}\int_0^P \rho V_+\overline{V_-}\,ds=\int_0^P p|V_0|^2\,ds. \tag{7}$$

For finite Fourier sums this follows because symmetrizing the coefficient $e^{d((n+\theta)-(m+\theta))}\rho_{n-m}$ replaces it by $\cosh(d(n-m))\rho_{n-m}=p_{n-m}$. General functions follow by Fourier truncation: both edge traces converge in L^2, and the middle trace also converges in L^2. Notice that (7) is independent of the Bloch phase. The real-part operation in (7) is essential: the unsymmetrized cross integral need not be real for complex data or a non-even weight. No evenness or reality of the Bloch function is assumed.

2. An inner factor with any prescribed nontrivial character

Fix $s_0\in\mathbb{R}$ and $y\in(-d,d)$, and put

$$c=\frac{\pi}{4d},\qquad \tau=\tan(cy),\qquad b_y(z)=\frac{\tanh(cz)-i\tau}{1+i\tau\tanh(cz)}.$$

The map $\tanh(cz)$ maps $\mathcal{D}_d$ conformally to the unit disk; b_y is its disk Blaschke factor for the point iy. Therefore $|b_y|<1$ inside, $|b_y|=1$ on the two edges, and its only zero in the strip is iy, a simple zero. Its limits as $\operatorname{Re} z$ tends to plus or minus infinity are

$$b_+=e^{-2icy},\qquad b_-=-e^{2icy}.$$

Define

$$B_y(z)=b_y(z-s_0)\prod_{j=1}^{\infty}\frac{b_y(z-s_0-jP)}{b_-}\,\frac{b_y(z-s_0+jP)}{b_+}. \tag{8}$$

Each normalized factor differs from one by an exponentially decaying quantity as $j \to \infty$, uniformly on a closed period rectangle and on a small neighborhood of its two horizontal edges. The product and its derivatives thus converge normally there. It defines a holomorphic function on the whole strip, smooth through the edges, whose zeros are exactly $s_0 + jP + iy$, all simple. Every factor has modulus at most one in the closed strip, so

$$|B_y| \le 1 \text{ in } \mathcal{D}_d, \qquad |B_y| = 1 \text{ on its two edges.} \tag{9}$$

More explicitly, fix $0 < \eta < d - |y|$. The nearest poles of a single factor are at heights $2d - y$ and $-2d - y$, outside $|\operatorname{Im} z| \le d + \eta$. On a bounded horizontal interval the normalized tails are $1 + O(e^{-2cjP})$, uniformly in that enlarged strip. Their deviations from one are summable. The infinite tail therefore has a nonzero limit away from the listed zeros, and normal convergence on a slightly larger compact set also gives convergence of derivatives. Thus the assertion about the exact zero set does not follow merely from boundedness of the finite products. Reindexing a finite symmetric product and taking its limit gives

$$\frac{B_y(z+P)}{B_y(z)} = \frac{b_+}{b_-} = \exp\Big(i\pi\left(1 - \frac{y}{d}\right)\Big). \tag{10}$$

The identity extends through the zeros. Hence its character parameter is

$$\chi(y) = \frac{1 - y/d}{2} \in (0, 1). \tag{11}$$

In particular every nontrivial unitary character occurs. Finally $\log|B_y|$ vanishes on the edges and is negative immediately inside; its outward normal derivative there is nonnegative. This follows already by taking the one-sided derivative of a smooth function that is nonpositive on an inward normal segment and zero at its endpoint. Write

$$\beta_\pm(s) = \partial_\nu \log|B_y(s \pm id)| \ge 0. \tag{12}$$

The product is character-automorphic, not generally periodic. No uniform edge regularity as $y \to \pm d$ is asserted or needed.

3. Division decreases energy and increases mass

For the main argument it suffices to take a finite Bloch Fourier series u, so that V in (3) is entire. Suppose it vanishes at $a = s_0 + iy$. By its character it vanishes at all $a + jP$. Consequently

$$W = V/B_y \tag{13}$$

is holomorphic throughout the strip and has character parameter $\theta - \chi(y)$ modulo integers. The apparent poles are removable, and B_y is analytic and nonzero through the two horizontal edges. Thus W is smooth through the closed doubled cell. It represents an admissible top trace by (4); no assertion about division of a rough trace is needed. Reconstruction is injective: on each real Bloch frequency it multiplies the middle Fourier coefficient by $\cosh(d(m + \theta - \chi(y))) > 0$, after an integer relabeling if needed.

The equal edge moduli and (6), together with $p > 0$ and $|B_y| \le 1$, give

$$M(V) \le M(W). \tag{14}$$

For the energy, the exact identity is

$$Q(V) - Q(W) = \frac{1}{4} \sum_{\pm} \int_0^P \beta_{\pm}(s) |W_{\pm}(s)|^2 \, ds \geq 0. \tag{15}$$

Indeed, for a holomorphic smooth function F, $\Delta |F|^2 = 4|F'|^2$, and hence (5) is one eighth of the boundary integral of $\partial_\nu |F|^2$. The lateral contributions cancel because $|F|^2$ is periodic. On the horizontal edges $|B_y| = 1$ and

$$\partial_\nu |B_y W|^2 - \partial_\nu |W|^2 = 2\beta_{\pm} |W|^2.$$

This proves (15) for precisely the smooth functions used below.

Combining (14)–(15) yields

$$\frac{Q(W)}{M(W)} \leq \frac{Q(V)}{M(V)} \quad (V \neq 0). \tag{16}$$

The division map is complex linear and injective on its domain.

4. All-phase ordered interlacing on the weighted strip

Let $\lambda_n(\theta)$ denote the ordered weighted strip eigenvalues. Fix arbitrary θ_a, θ_b. If their characters coincide, $\lambda_n(\theta_a) \leq \lambda_{n+1}(\theta_b)$ is the ordinary ordering. Otherwise choose y so that

$$\chi(y) = \theta_b - \theta_a \pmod 1, \qquad 0 < \chi(y) < 1. \tag{17}$$

Fix $\varepsilon > 0$. Approximate a basis of the first $n+2$ eigenfunctions in phase θ_b by finite Bloch Fourier series in $H^{1/2}$. Since the form and mass Gram matrices converge, with the mass matrix remaining positive definite, this gives an $(n+2)$-dimensional finite Fourier subspace L such that

$$\max_{0 \neq u \in L} \frac{q_{\theta_b}[u]}{\int \rho |u|^2} \leq \lambda_{n+1}(\theta_b) + \varepsilon. \tag{17a}$$

This remains true when eigenvalues have multiplicity: any orthonormal eigenbasis for their span can be approximated in this way. Evaluation $V \mapsto V(s_0 + iy)$ is one complex linear functional on L. Its kernel intersection has dimension at least $n+1$. Apply division (13) and reconstruct top traces by (4). This gives an injective linear image of dimension at least $n+1$ in phase θ_a, and (16) holds for every nonzero vector. The min–max principle consequently gives

$$\lambda_n(\theta_a) \leq \lambda_{n+1}(\theta_b) + \varepsilon.$$

More explicitly, the maximum quotient on L is at least the maximum on its kernel intersection, which is at least the maximum on the divided image, which is at least $\lambda_n(\theta_a)$. Letting $\varepsilon \downarrow 0$ proves

$$\lambda_n(\theta_a) \leq \lambda_{n+1}(\theta_b). \tag{18}$$

The approximation is only in one finite-dimensional spectral subspace for fixed n and phase. No uniform-in-index approximation or division of a general form-domain function is being assumed.

5. Exact geometric reduction for analytic graph bottoms

Suppose first that b is real analytic and periodic, with positive depth $H(x) = h - b(x)$. The physical channel modulo horizontal period is conformally an annulus. For example $\zeta = e^{-iz}$ identifies it with the doubly connected planar domain bounded by the unit circle and the analytic simple curve $\zeta = e^{-H(x)}e^{-ix}$; its two boundary components are disjoint. Annular uniformization, with its analytic boundary extension, gives after lifting logarithms a conformal map

$$Z : \{-d < \operatorname{Im} w < 0\} \longrightarrow \Omega_b, \qquad Z(w + P) = Z(w) + P, \tag{19}$$

for some $d > 0$, taking top to top and bottom to bottom. The map and its inverse extend analytically locally through either boundary, and their derivatives are nonzero there. The horizontal period is still exactly P; no depth-to-period ratio is discarded.

For precise standard references, annular uniformization is the connectivity-two case of L. V. Ahlfors, *Complex Analysis*, third edition, McGraw–Hill, 1979, Chapter VI, Section 5.1, Theorem 10, pp. 255–256. The analytic boundary extension is the local reflection argument in Chapter VI, Sections 1.3–1.4, Theorems 3–4, pp. 233–235; locally straightening an analytic boundary arc and a circular annulus boundary gives the same reflection situation. The required hypotheses here are analytic embedded boundary circles separated by a positive distance.

On the top $Z(s)$ is real and strictly increasing, so

$$\rho(s) = Z'(s) > 0.$$

On the bottom write $Z(s - id) = X(s) + i(-h + b(X(s)))$. This parametrizes the graph once with the positive horizontal orientation, so

$$p(s) = \operatorname{Re} Z'(s - id) = X'(s) > 0. \tag{20}$$

For completeness, X' cannot vanish: the derivative of a regular parametrization of a C^1 graph is proportional to $(1, b'(X))$, and the conformal derivative is nonzero. Orientation and degree one fix its positive sign. The periodic holomorphic function Z' is real on the top. Its Fourier series therefore has conjugate-symmetric coefficients, and taking its real part at depth d gives

$$p_k = \cosh(dk)\rho_k, \tag{21}$$

exactly (2). Both weights are continuous and strictly positive.

Conformal invariance of the two-dimensional Dirichlet integral transforms the physical harmonic minimization into the flat-strip minimization. The natural bottom condition remains zero Neumann. The physical surface measure becomes $dx = \rho(s)ds$, and (19) preserves the Bloch character exactly. The image of a conformal period cell may have curved lateral sides; the energy and mass densities are periodic, so integration over this fundamental cell agrees with integration over the physical vertical-sided cell. Thus the physical spectrum is precisely the weighted strip spectrum used in (18). This proves (1) for analytic graph bottoms, without amplitude, parity, fundamental-period or genericity assumptions.

6. Passage to every admissible C^1 graph

Choose real trigonometric polynomials $b_j \to b$ in C^1 and keep a common positive lower depth. With $H_j = h - b_j$ and $H = h - b$, pull the physical bulk energies back by $y = H_j(x)z$ and $y = H(x)z$ to

the fixed cell $(0,P)\times(-1,0)$. Their coefficient matrices are

$$A_H(x,z)=\begin{pmatrix} H & -zH' \\ -zH' & (1+z^2(H')^2)/H \end{pmatrix}, \tag{22}$$

acting on $((\partial_x+i\theta)U, U_z)$. Uniform positivity of the depths and C^1 convergence imply uniform ellipticity and uniform convergence of these matrices. Consequently, for numbers $\varepsilon_j\to 0$,

$$(1-\varepsilon_j)q_{b,\theta}\le q_{b_j,\theta}\le(1+\varepsilon_j)q_{b,\theta}, \tag{23}$$

uniformly in phase. This follows first for every bulk competitor and then by taking the infimum with its fixed top trace. Min–max therefore gives convergence of every fixed ordered eigenvalue, including its multiplicities. For fixed n,θ_a,θ_b, pass (1) for b_j to the limit using (23). This proves (1) for b. Since the phase pair was arbitrary, it gives the asserted max–min inequality.

For clarity, phase continuity follows directly from that same bulk realization. If u is the top trace of U and $d_+=\max H$, vertical integration gives

$$\|U\|^2_{L^2(\mathcal{C}_b)}\le 2d_+\|u\|^2_{L^2}+2d_+^2\|U_y\|^2_{L^2(\mathcal{C}_b)}.$$

Writing $a_\theta[U]$ for the bulk energy, expansion of $|(\partial_x+i\theta+i\delta)U|^2$ and Cauchy–Schwarz therefore give, for $|\delta|\le 1$,

$$|a_{\theta+\delta}[U]-a_\theta[U]|\le C|\delta|\big(a_\theta[U]+\|u\|^2_{L^2}\big),$$

where C depends only on d_+. Apply the upper bound to the minimizing extension at θ, and interchange the two phases for the reverse bound. The resulting two-sided comparison of $q_\theta+\|u\|^2$ and $q_{\theta+\delta}+\|u\|^2$, followed by min–max, proves continuity of each fixed ordered eigenvalue. Compactness of the phase interval gives its maximum and minimum.

The functional-analytic ingredients are the representation of a densely defined closed nonnegative form by its self-adjoint operator and its compact-resolvent min–max principle; the form domain here is $H^{1/2}$ on the top circle and its embedding in L^2 is compact. Standard form background is T. Kato, *Perturbation Theory for Linear Operators*, second edition, corrected printing, Springer, 1980, Chapter VI, Theorem 2.1. All phase-changing comparisons in the proof were established explicitly by a single analytic evaluation constraint and inner factor division. No finite-rank assertion about a vertical cross-section, or oscillation theorem for real Bloch functions, is used.